Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Feature Engineering before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Feature Engineering trace and record the first step you can explain without help.

Use W2: Transform validation for **Feature engineering**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Feature Engineering explanation against the supplied facts before discussing it.

Use the worked case for **Feature engineering**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Feature Engineering target to a trace and a justified conclusion.

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Feature Engineering case and show the deciding evidence.

Features are leaf length and width in cm; add length/width.

Compute for8 and2; state boundary checks.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 29 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Feature Engineering adds to Responsible Plant Classifier and keep the evidence.

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Compute features

> Compute features. Apply the supplied Feature Engineering facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Feature Engineering", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Compute the worked areas and their **training** mean.

> **My reasoning and evidence**

WORKBOOK · 13 OF 26

W2 Transform validation

> Transform validation. Apply the supplied Feature Engineering facts and keep your reasoning.

Center the validation area using the frozen **training** mean.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Reject leakage

Reject leakage. Apply the supplied Feature Engineering facts and keep your reasoning.

Why is delivery delay invalid for a **prediction** made before dispatch?

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Feature Engineering facts and keep your reasoning.

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Plan a fair comparison

Plan a fair comparison. Apply the supplied Feature Engineering facts and keep your reasoning.

How would you test adding area without claiming it must help?

My reasoning and evidence

WORKBOOK · 17 OF 26

Relationships and confounding — W1 Describe the evidence

> Use the supplied Feature Engineering task to record your reasoning, evidence and next step.

State the pattern in the three worked records without using **cause**d or guaranteed.

My reasoning and evidence

Relationships and confounding — W2 Identify a confounder

> Use the supplied Feature Engineering task to record your reasoning, evidence and next step.

Explain how prior experience could relate to both practice time and score.

My reasoning and evidence

Relationships and confounding — W3 Repair an overclaim

> Use the supplied Feature Engineering task to record your reasoning, evidence and next step.

A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

My reasoning and evidence

Relationships and confounding — W4 Plan independently

> Use the supplied Feature Engineering task to record your reasoning, evidence and next step.

A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer **comparison** of whether hints help.

My reasoning and evidence

Relationships and confounding — W5 State a remaining limit

> Use the supplied Feature Engineering task to record your reasoning, evidence and next step.

> Even a better **comparison** has only four fictional groups. What should be reported?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Feature Engineering

> Engineer a feature with a transformation defined before evaluating a new leaf. Explain the labelled step before checking the changed condition.

Engineer a feature with a **transformation** defined before evaluating a new leaf.

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/29

Feature Engineering

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Construct area

Query area=2 x 4=8 cm2.



2. Fit centering

Training mean=(2+6+10)/3=6 cm2.



3. Apply unchanged transform

Centered query feature=8-6=2 cm2.

CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Engineer a feature with a **transformation** defined before evaluating a new leaf.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Centered query feature=8-6=2 cm².

> 1. Construct area

> 2. Fit centering

3. Apply unchanged transform

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply unchanged transform Centered query feature=8-6=2 cm². Read the picture in order. Centered query feature=8-6=2 cm².

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Include the query in the centering mean. Which conclusion is supported by the supplied facts?

1. Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Repair a missing building block

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What **comparison** is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

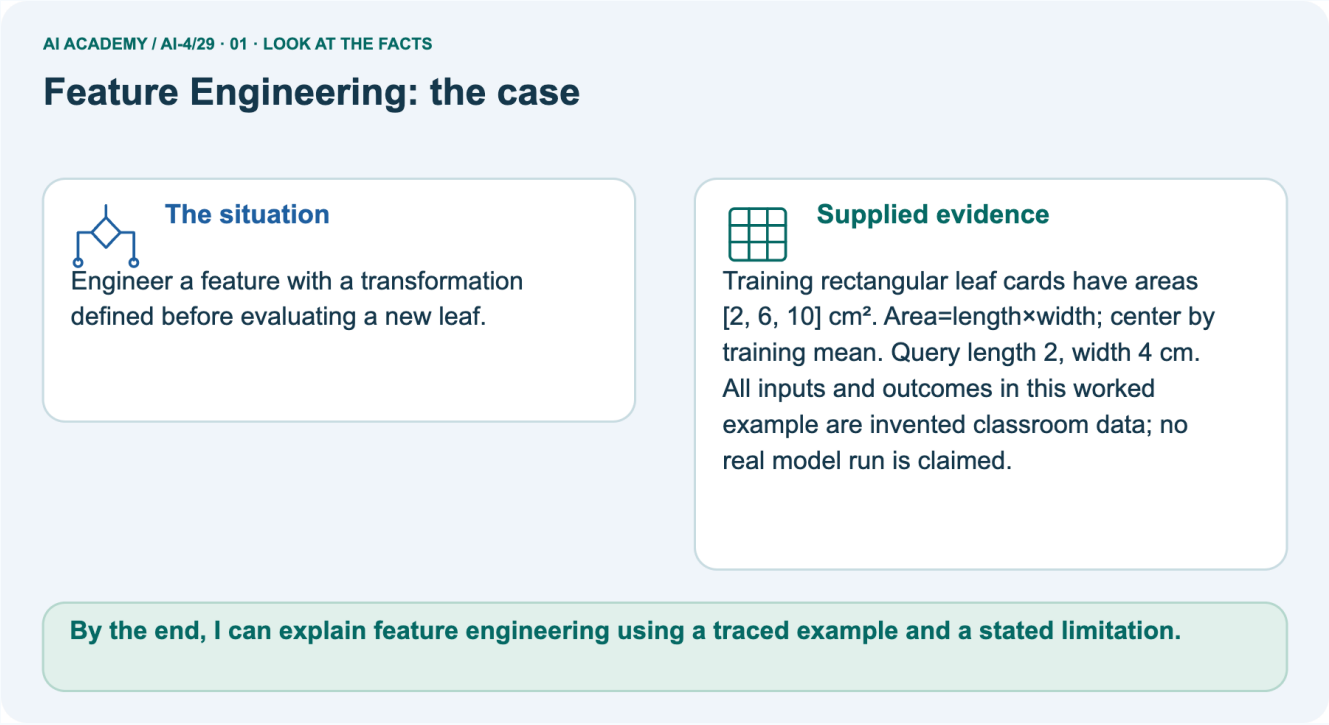
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Feature Engineering picture from its labels, then record a relevant limit.

01. Feature Engineering: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

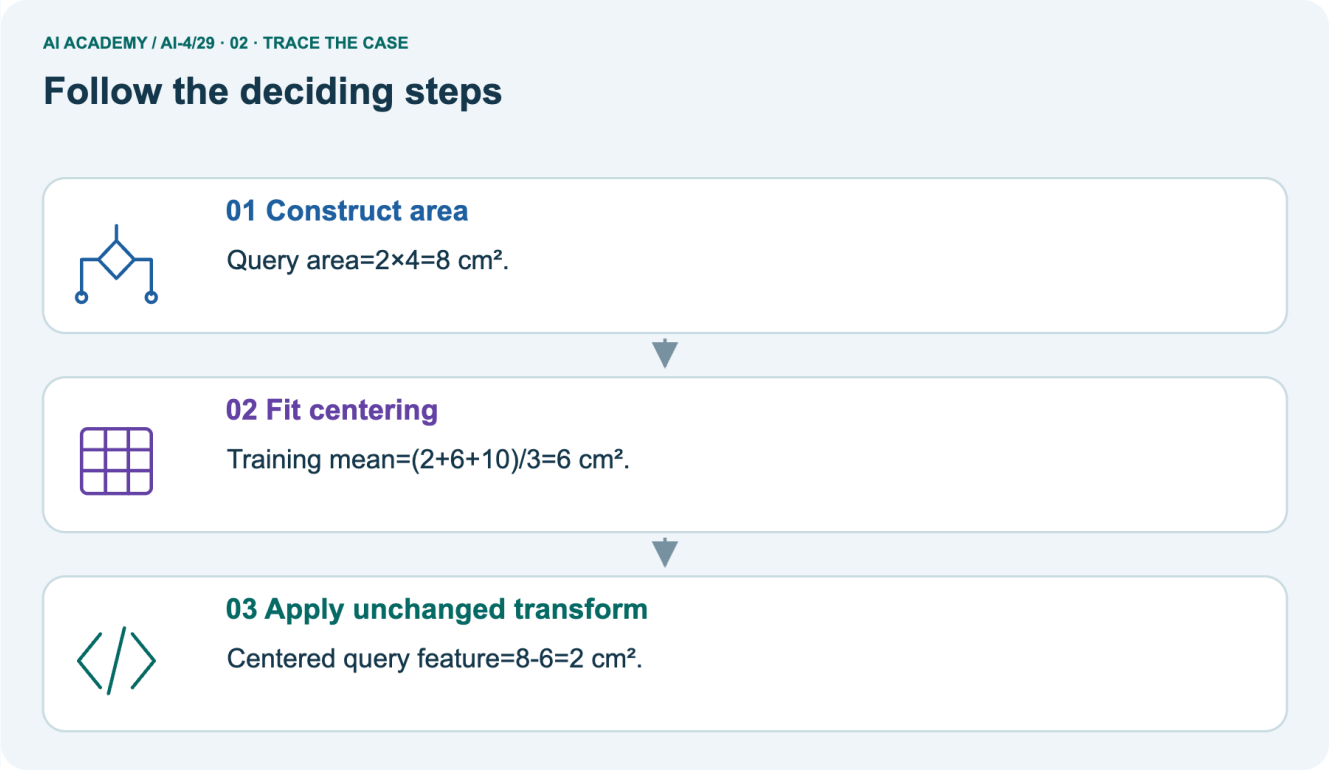
Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



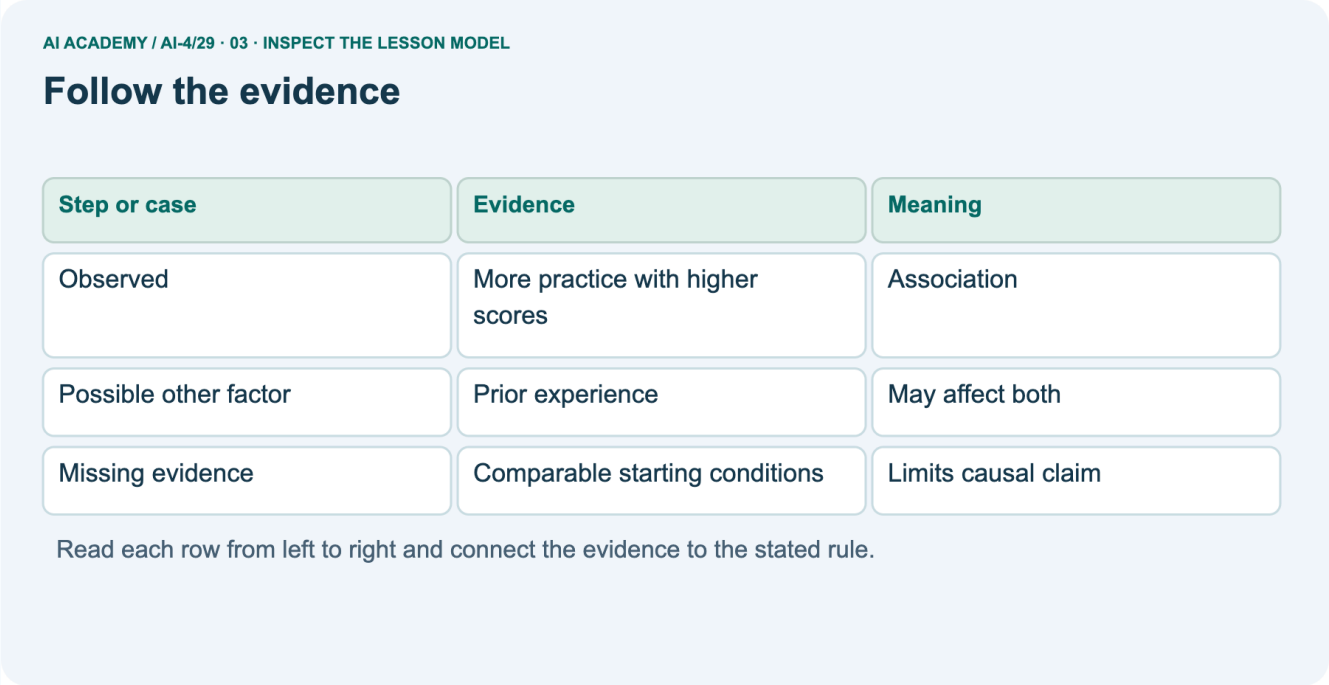
Follow the labelled picture with the matching explanation. Open the vector diagram

Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm². → Apply unchanged transform: Centered query feature=8-6=2 cm².

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



AI ACADEMY / AI-4/29 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Observed	More practice with higher scores	Association
Possible other factor	Prior experience	May affect both
Missing evidence	Comparable starting conditions	Limits causal claim

Read each row from left to right and connect the evidence to the stated rule.

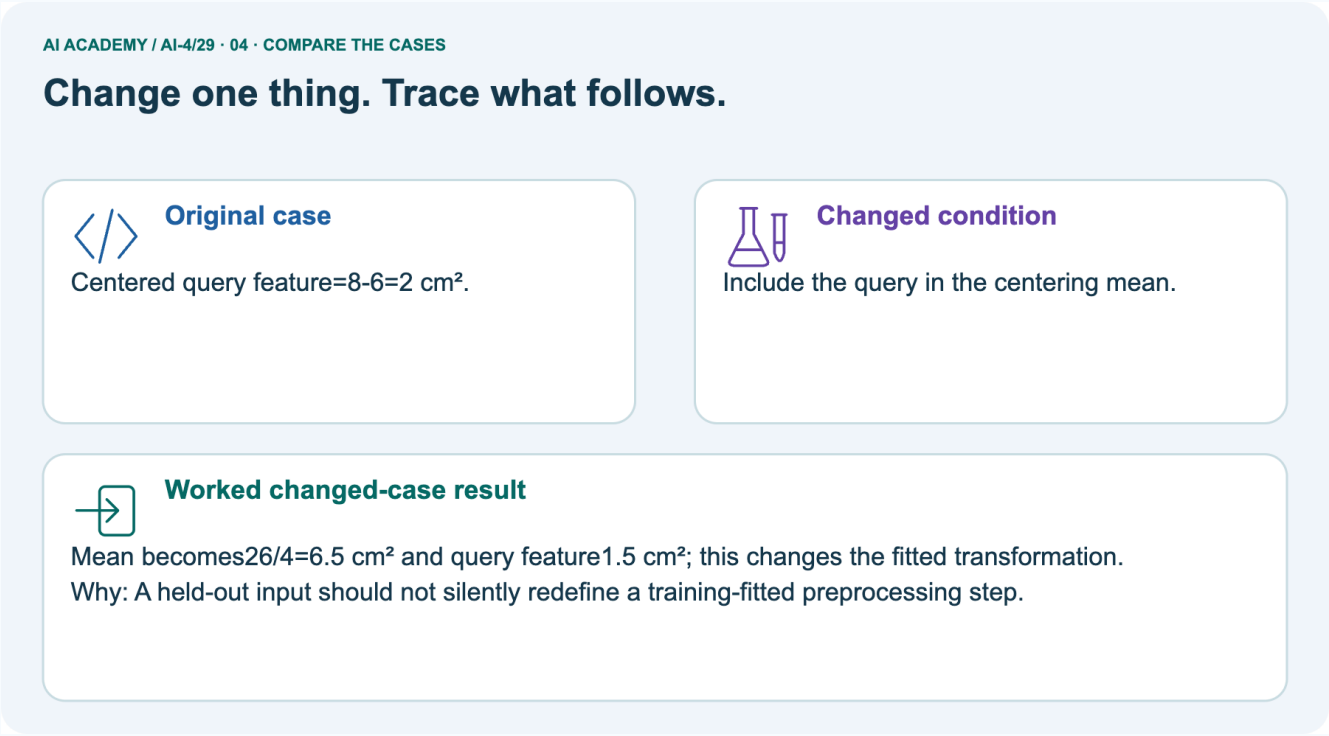
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

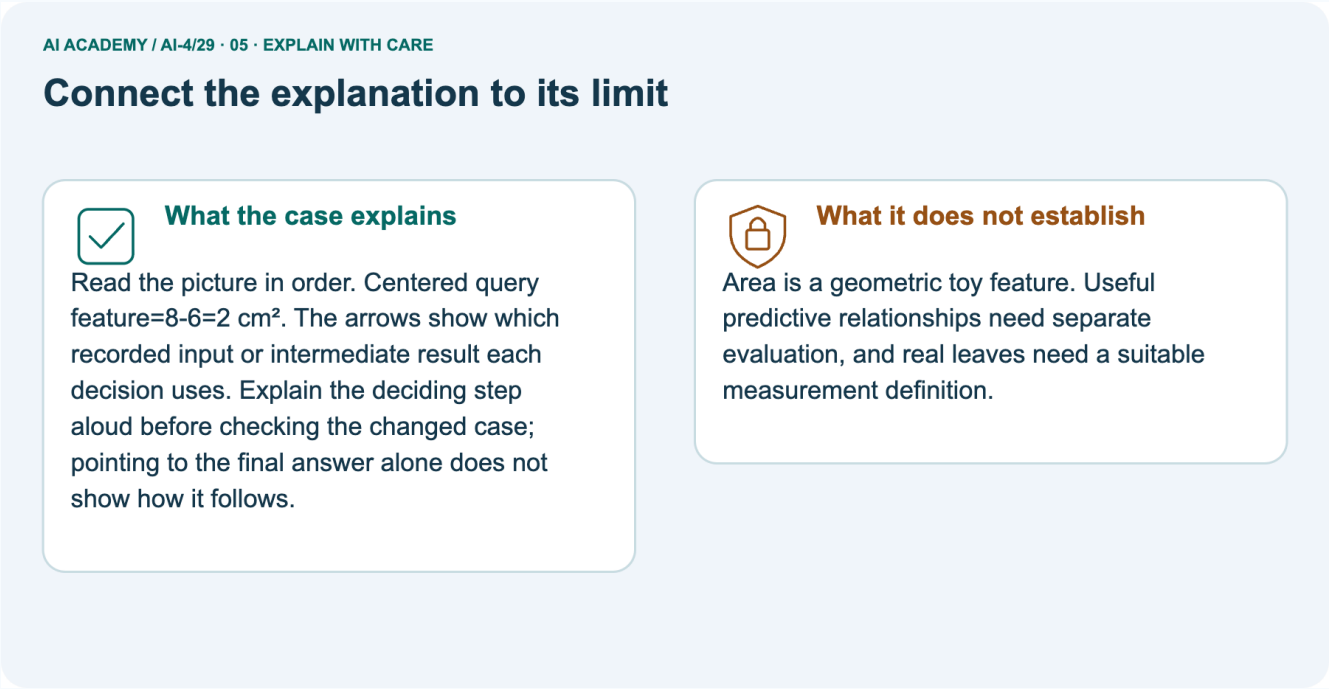
Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

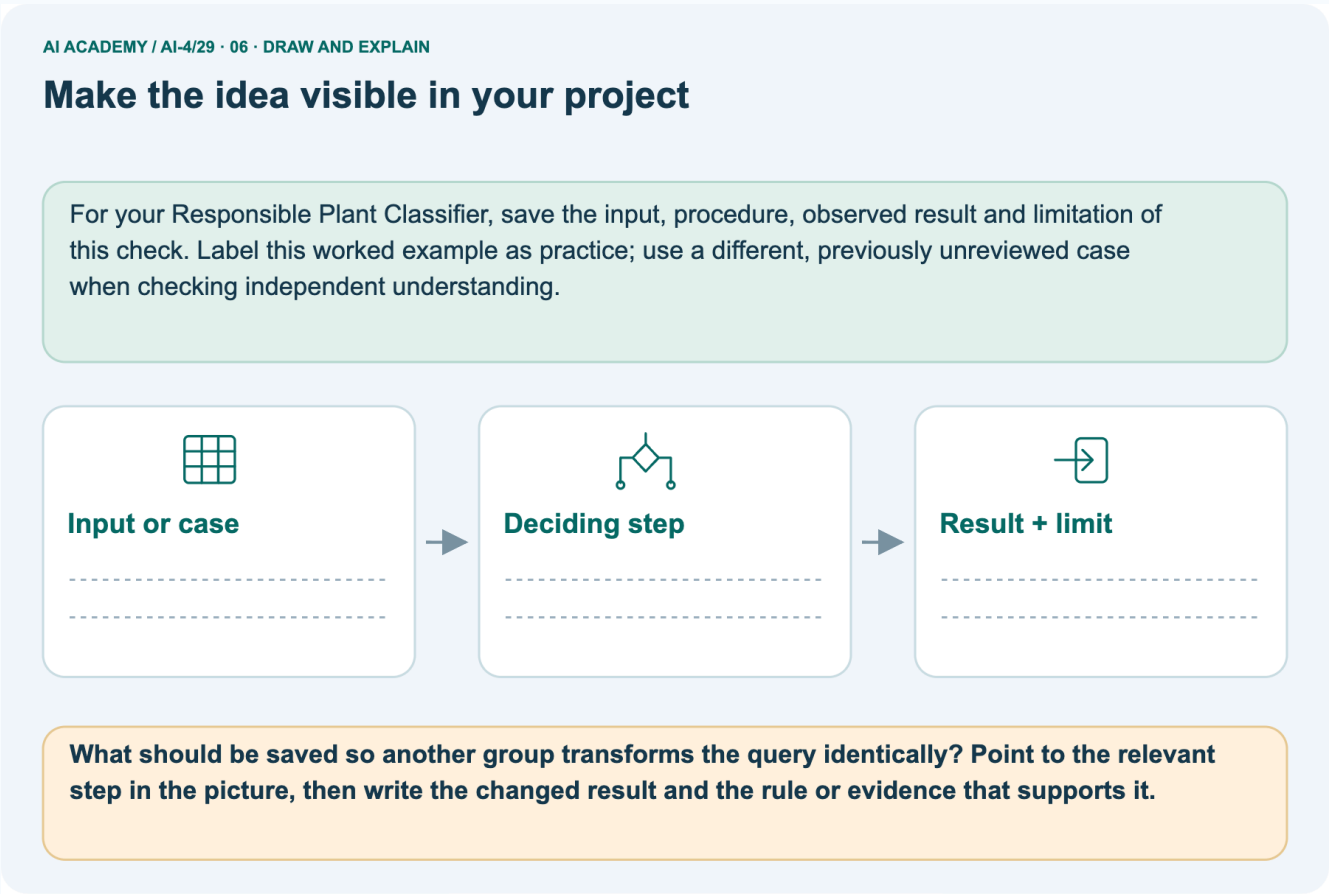
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.